Precision-fermented whey and egg proteins: what they cost, and where funding helps

Memo · draft v1 (2026-09-26 15:19) · Pablo Antonio Moreno Casares · techno-economic model `pf-tea`, Humbird (2021) method, calibrated against NREL, Intelligen and Voutilainen designs

Bottom line.

- **Today's public strains are an order of magnitude too expensive.** Food-grade β-lactoglobulin (BLG, whey) and ovalbumin (OVA, egg) cost $188 and $467 per kg protein at the best published titers. Whey protein trades at $7–$18 and dried egg white at $8–$20 per kg protein (2015–2025).
- **Industry-level strains are close but not there.** At the 42 g/L private-sector average titer, the cost is $22.8 (BLG) with a conservative capital estimate and $13.3 with the lighter one used by published studies. That undercuts whey only in spike years like 2026 (about $34/kg protein), or, with light capital, at mid-range whey prices.
- **Physics caps the titer, and capital dominates the cost.** Oxygen transfer and cooling limit a 200 m³ fermenter to about 54 g/L in a 72 h production phase. Capital-related costs are 68% of the base-case price; sugar is 17%.
- **The physical floor is low.** With everything at its best at once, the cost is $6.8 (BLG) and $11.4 (OVA) per kg, or $3.5 and $6.3 with light capital. The gap between today and the floor is engineering and biology, not thermodynamics.

$/kg protein, 10 kt/yr, US	titer g/L	BLG	BLG light	OVA	OVA light
best public academic strain	5 / 2	$188	$103	$467	$230
base case	30	$29.2	$16.8	$31.6	$18.4
industry benchmark	42	$22.8	$13.3	$25.2	$14.9
physical limit (100 kt/yr)	130 / 50	$6.8	$3.5	$11.4	$6.3

What $20/kg requires. About 38 g/L of BLG in a 48 h production phase, or 46 g/L in 72 h, with conservative capital. That is at the private-sector average and near the oxygen ceiling. With light capital, 21–25 g/L. $10/kg is not reachable with conservative capital at today's cell densities and yields. With light capital it needs 62–69 g/L in 4–5 days, right at the oxygen ceiling. Beyond that, cell density, production time, cheaper capital and scale must move together. Product yield and vessel size, perhaps surprisingly, matter little.

Where the next dollar helps most, in our judgment:

1. **Open fed-batch data.** Measured titer, productivity and sugar yield **together**, at 1–10 m³, for BLG and OVA. No public dataset has all three. This is the largest evidence gap, and every cost estimate, ours included, rests on company averages.
2. **Productivity per hour of fermenter time, not titer alone.** Near the oxygen ceiling, extra titer helps only if it does not take longer.
3. **Capital-light capacity.** Capital is the largest cost item in every scenario. Conversions of existing ethanol or enzyme fermenters, and shared scale-up plants, attack it directly.
4. **Downstream data.** Membrane fluxes and recoveries at scale (today: one patent example) are the next largest uncertainty.

EU. At EU sugar and energy prices the same plant costs about $4.4/kg more, mostly because of sugar. The cost of capital matters more than energy subsidies.

Caveats. A production cost with a 14.6%/yr capital charge, not a market price. Excludes R&D, licensing, regulatory, sales and first-of-a-kind premiums. Strain parameters rest on design values and company-reported averages (evidence grades B–C). OVA results depend on an assumed 50 g/L practical solubility. Full report, calibration record and data: `pf-tea` repository.